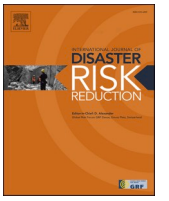
International Journal of Disaster Risk Reduction 127 (2025) 105669



Contents lists available at ScienceDirect

International Journal of Disaster Risk Reduction

journal homepage: www.elsevier.com/locate/ijdrr





Uncovering compound urban crises with large language model-assisted knowledge graph construction

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A B S T R A C T

Urban areas frequently face an array of crises, including natural hazards, public health emergencies, and technological disruptions, which are becoming more interconnected due to the projected growing trend of climate change and rapid urbanization. While prior studies have investigated compound events, much of the focus has been on weather- and climate-related events, leaving a gap in understanding compound urban crises that integrate diverse types of crises at the city level. To address this, we propose a novel method for mining location-specific knowledge of compound urban crises using large language models (LLMs), enhanced with techniques such as Retrieval-Augmented Generation (RAG), iterative self-refinement, and prompt engineering. By applying the method to 1941 news articles collected from Shenzhen, China, we constructed knowledge graphs (KGs) that reveal the interconnections and compounding patterns among 13 distinct types of urban crises. The results demonstrate the effectiveness of the proposed method in identifying and mapping compound urban crises at the city level. The findings offer practical implications for urban crisis management, equipping cities with tailored knowledge to better anticipate and respond to complex, interrelated crisis scenarios.

1. Introduction

Cities, as vital components of modern society, frequently face various public crises, including extreme weather events, public health emergencies, social unrest, and technological disruptions. These crises pose significant threats to urban functionality, economic stability, and social cohesion [1]. Moreover, the frequency and complexity of urban crises are expected to increase with the projected trends of global warming, climate change, and rapid urbanization. Importantly, these crises often do not occur in isolation or affect separate regions at different times; instead, they are increasingly interlinked, with one event influencing or amplifying another [2]. For example, a heatwave can exacerbate air pollution by increasing ground-level ozone concentrations while simultaneously straining public health systems during a pandemic, leading to compounded mortality and morbidity rates. Such interconnected events, known as *compound events*, have garnered growing attention in recent research, particularly in climate and environmental science (e.g., Ref. [3–5]).

The concept of *compound events* differs from related ideas such as *disaster chains* and *cascading failures*. While disaster chains describe a sequence of events triggered by an initial disaster (e.g., an earthquake leading to tsunamis and infrastructure collapse), and cascading failures emphasize the propagation of failures across interconnected systems (e.g., power grid failures disrupting communication networks), compound events focus on the concurrent or consecutive interaction of multiple hazards that collectively amplify societal impacts [3,6,7]. For instance, the interplay between a hurricane and subsequent flooding can create a compound crisis that overwhelms emergency response systems, highlighting the need for a distinct framework to address such complexities.

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https://doi.org/10.1016/j.ijdrr.2025.105669
Received 17 February 2025; Received in revised form 18 May 2025; Accepted 25 June 2025
Available online 26 June 2025
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Despite these advancements, existing studies on compound events have predominantly centered on extreme climate and weather phenomena, relying heavily on scientific projections or historical meteorological data (e.g., Ref. [5,8,9]). While valuable, these studies have largely overlooked other types of crises, such as industrial accidents, pollution events, or public health emergencies, that may interact with climate-related events to produce more devastating impacts. Moreover, although research has acknowledged that the risks of compound crises vary across geographic contexts, there is limited knowledge specific to the city level. Cities, as the primary actors in crisis management, require tailored strategies to address the unique geographic and social contexts of compound events.

One pioneering study in this area is Westman et al. [10], which expanded the concept of compound events to urban contexts and introduced the notion of *compound urban crisis*. Drawing on ideas from complex adaptive systems and critical urban studies, Westman et al. [10] developed a set of boundary concepts—unsettlement, unevenness, and unbounding—to explore the complexities of compound urban crises. While this study offers valuable theoretical insights and illustrative case examples, it does not provide a systematic mapping of compound urban crises for a city from a risk management perspective. Mapping and understanding the compounding patterns of urban crises can be challenging [10]. First, urban crises encompass a wider range of hazardous events, such as public health crises and security accidents, in addition to climate and weather-related events, which add complexity to the interactions among events. Second, the drivers of urban crises can extend beyond natural systems to include social and built environment factors, creating a highly intricate web of causality. Additionally, many urban crises (e.g., building fires, noise pollution) unfold at finer spatial scales and lack systematic sensor-based data collection, unlike climate and weather events. Public agencies may track major events, but these records are often unstructured, incomplete, or unavailable, exacerbating the difficulty of identifying and managing compound urban crises effectively.

Recent advancements in artificial intelligence (AI) offer promising opportunities to address these challenges. Some recent studies have proposed AI-based approaches for mining useful information from unstructured data (e.g., Ref. [11–13]). Particularly, knowledge graphs (KGs) have shown great potential for capturing the complex and evolving relationships among sub-events involved in disasters (e.g., Ref. [1,14]). In this study, we propose a method to mine localized knowledge on compound urban crises by leveraging the advancements in Large Language Models (LLMs) and integrating techniques, such as Retrieval-Augmented Generation (RAG), iterative self-refinement, and prompt engineering, to ensure the accuracy and completeness of the model output. The method processes unstructured textual reports of historical events (e.g., news articles) and generates KGs as output, with entities representing different types of events and edges depicting their relationships. We applied this method to 1,941 news articles covering 13 types of crisis events collected from Shenzhen, China. The analysis demonstrates that the proposed method effectively identifies diverse compound events, revealing intricate interconnections among crises. This method not only supplements conventional models for studying compound climate events but also provides location-specific insights to support urban crisis management, helping cities better prepare for the complexities and challenges of compound urban crises.

2. Literature review

2.1. Theoretical and empirical studies on compound events

The concept of *compound events* was first introduced by the Intergovernmental Panel on Climate Change (IPCC) Special Report on Climate Extremes in 2012, which was further expanded and supplemented by many researchers [3,5,15,16]. Generally, it was referred to as the "*combinations of multiple climate drivers and/or hazards that contribute to societal or environmental risk*" [3,5,15]. The hazards and drivers can be either of the same type, e.g., sequential storms attacking the same region, or of different types. Also, the events can be either independent - they occur due to randomness – or mutually correlated, e.g., the rain storm can increase the probabilities of flooding and landslides. Compared to individual hazardous events, the coupling of multiple hazards is more likely to generate non-linear and cascading impacts and is associated with more severe disaster impacts.

Some researchers have developed typologies to categorize compound events referring to the spatial, temporal, and causal relations among the different types of consisting events (e.g., Ref. [17,18]). Particularly, Zscheischler et al. [4] summarized four general types of compound extreme events referring to the relations among four key characteristics, i.e., drivers, hazards, impacts, and modulators, where *drivers* are climate processes; *hazards* refer to the climate-related phenomenon before a potential impact (the proximate cause of the impact); *impacts* are the outcomes of the events; and *modulators* influence the frequency and location of *drivers*, and, thereby, the frequency and/or intensity of *hazards*. The four types of compound events summarized by Zscheischler et al. [4] are: *preconditioned events* that a weather-driven or climate-driven precondition aggravates the impacts of a hazard; *multivariate events* that multiple drivers and/or hazards lead to an impact; *temporally compounding events* that a succession of hazards leads to an impact; and *spatially compounding events* that spatially co-occurring hazards cause an impact.

While typology development provides a theoretical framework for researchers to understand and classify compound events, some studies have also developed numeric approaches to analyze compound events and uncover the relationships among sub-events. For example, some researchers leveraged spatio-temporal correlation-based approaches to study the joint occurrence of multiple hazards, referring to historical meteorological observations, and mapped the hotspots of compound events for different countries (e.g., Ref. [8,9]). Considering the complex coupling and correlation between the different hazardous variables, some scholars proposed to use graph-based approaches, such as Bayesian networks, spatio-temporal networks, and complex network models, to model the dependencies among different variables (e.g., Ref. [16]). Additionally, Copula is widely used to map the multivariate probability distributions of different hazard variables and was coupled with other modeling approaches such as the Bayesian network (e.g., Ref. [16,19]).

In addition to the numeric approaches developed to map the risks of different types of compound events, researchers have

conducted specific case studies to supplement empirical evidence of compound events across the world. For example, Hao [5] summarized four common types of compound events in China, including compound heatwaves, heatwave-droughts, flood--rainstorm-heatwaves, and compound flooding events. Zong et al. [8] investigated the occurrence and hotspots of multivariate and temporally compounding events in China using historical meteorological observations. Their findings revealed that regions such as the Yangtze River Basin and southeastern China are particularly prone to compound heatwaves and concurrent precipitation extremes. Beyond China, Tavakol et al. [19] explored the probability of compound climate extremes in the central U.S., focusing on hot, dry, and windy events. The copula-based analysis showed that the likelihood of such compound events has increased under climate change, posing significant risks to agriculture, water resources, and wildfire management. Chen et al. [9] examined compound wet and dry extremes, showing that events, such as concurrent droughts and floods, are becoming more frequent in certain areas.

Though significant, previous studies on compound events are limited to climate and weather events. However, some recent studies have indicated that those climate and weather events can also correlate with urban crises, such as pollution, pandemics, urban fires, power outages, and traffic jams, to exert cascading impacts on human societies, which is referred to as "compound urban crises" [10], demanding more attentions from urban emergency management practices. Secondly, previous research studying compound climatic/weather events mostly adopted a "top-down" approach and heavily relied on the historical or projected probabilities of different climatic and meteorological processes. Compound events can involve different types of hazards, which can occur in different temporal sequences (e.g., simultaneously or sequentially), making them more challenging to study with conventional approaches. Therefore, many studies only focused on a particular type of compound events, such as compound flooding, heatwave, and droughts (e.g., Ref. [9, 19]), except that Zong et al. [8] compared 44 types of possible compound events in China with a complete set of meteorological station readings across a 60-year period. Instead of this "top-down" approach, Leonard et al. [16] have suggested adopting a stakeholder-centered approach by understanding the information required for local stakeholders to make decisions. Moreover, previous studies have indicated that the risks of compound events are highly place-specific, i.e., can vary across different cities. However, the mapping of compound event risks is often conducted in coarse regions and is less practical for urban risk management. Therefore, more location-aware knowledge of compound events is needed.

Given these presented challenges, we propose an approach leveraging LLMs to build city-level profiles for compound urban crises based on historical news articles reporting the different types of urban crisis events, including both climate/weather events and other types of urban crises.

2.2. The application of knowledge graphs in crisis management

The structured format of knowledge in KGs is machine-readable, making it effective for information management. One application of KGs in crisis management is facilitating knowledge sharing and collaborative decision-making among stakeholders during emergencies [1,20]. For example, researchers have developed KGs, taking inputs from government regulations and standards, to enable rapid information dissemination during crises [21,22]. This capability is particularly beneficial for on-site first responders, as KG-supported information queries can guide appropriate actions in complex and dynamic scenarios [21]. Liu et al. [14] constructed a KG to represent and store complex entities and relationships related to flooding events in Beijing, identifying critical nodes for effective flood information management. Similarly, Anjomshoaa et al. [23] proposed a KG representation of cities for crisis management, dividing urban areas into square grids as the basic entities of the KG. Each grid was associated with data on infrastructure elements, service networks, hazard vulnerability, and environmental indicators, providing a comprehensive framework for urban crisis management.

Another application of KG in crisis management is the empirical exploration of complex crisis processes. For example, some researchers have constructed a KG from vast amounts of social media data to track the evolving dynamics of flooding events (e.g., Ref. [24,25]). Jiang et al. [26] have developed and visualized a KG to show the spatiotemporal conditions of COVID-19 with multi-sourced data. Similarly, Ye et al. [27] proposed a KG framework for typhoon events, focusing on the evolving status (i.e., location and attributes) of typhoons. Beyond visualizing the spatiotemporal evolutions of crisis events, some studies have developed analytic approaches based on KG. For example, Daher et al. [20] constructed a KG for flooding events by integrating various community elements (e.g., population, facilities, buildings, vulnerability, and hazard) and used rule-based algorithms to assign evacuation priorities to different locations efficiently. Ge et al. [28] introduced a KG framework for predicting forest fires by incorporating heterogeneous spatiotemporal data, including topographic, vegetation, meteorological, and human activity factors. The use of KGs enables the integration of domain expertise and machine learning algorithms, represented with semantic rules, to improve predictive accuracy.

Many of these existing studies rely on human annotators to construct KGs (e.g., Ref. [26]), which can be labor-intensive and time-consuming. Recent advancements in NLP and LLMs have enabled the exploration of AI-assisted approaches for automated KG construction (e.g., Ref. [21,24,29]). Particularly, BERT-based approaches have demonstrated exceptional performance in Named Entity Recognition (NER), which can be leveraged to extract entities for KG construction (e.g., Ref. [24,30]). However, these methods often require a substantial amount of high-quality annotated data for model fine-tuning. LLMs can mitigate this limitation with their inherent general knowledge and can perform cognitive tasks like humans. A few recent studies have explored the application of LLMs in crisis management tasks, such as estimating disaster-impacted locations (e.g., Ref. [31]), assessing flood depth (e.g., Ref. [32]), and enabling fast information extraction (e.g., Ref. [22,33,34]). Some scholars have also examined the application of LLMs for KG construction, though not specifically focusing on crisis management (e.g., Ref. [21,29,35]). Though powerful, these LLM-assisted approaches are constrained by the knowledge within their training data and may produce inaccurate or "hallucinated" information [36]. This limitation is particularly relevant in urban crisis management, where crises are highly complex and involve diverse interlinked factors, complicating the identification of entities and relations for KG construction.

3. Case description and data collection

In this research, we demonstrate the proposed approach using data collected from Shenzhen, China. Several factors make Shenzhen an ideal case for investigating compound urban crises. First, as a coastal city in southern China, Shenzhen is highly vulnerable to several climate and weather hazards, including heatwaves, typhoons, and flooding, among others. Secondly, as one of the fastest-growing cities in the world, Shenzhen has experienced significant changes in infrastructure, population demographics, and environmental conditions, all of which have contributed to a complex web of interconnected challenges, ranging from traffic congestion to pollution. By studying Shenzhen's experiences, policymakers and urban planners can gain valuable insights into addressing similar issues in other rapidly growing cities, ultimately promoting sustainable and resilient urban development.

The research team collected data by crawling news articles from the Shenzhen News website (https://sznews.com), an official platform that archives news articles related to Shenzhen from a wide range of mainstream and regional news agencies, including government-affiliated and commercial media outlets. We focused on articles published over a three-year period, from January 1, 2021, to April 1, 2024. News articles were chosen as the primary data source for several reasons. First, they provide relatively comprehensive and publicly accessible records of real-world events. Almost all events that generate certain societal impacts will be reported by major or minor news agencies, ensuring broad coverage of relevant incidents. Second, news articles often include localized insights about events, such as discussions of their associations with surrounding social and built environment conditions. This contextual information is critical for understanding the unique geographic and social dimensions of compound urban crises. However, readers are free to explore the applicability of this method to other types of unstructured textual data, such as social media posts.

To identify news articles related to climate/weather hazards and diverse urban crises, the research team used a set of keywords to search for potentially qualified news articles, as listed in Table 1. Given the warm climate and geographic context in Shenzhen, we did not include events like snowstorms and earthquakes due to their rare occurrences. The initial search returned 4,697 qualified news articles, which further went through manual verification to remove news articles that introduce policy documents, guidelines, stories, treatments, or education campaigns for climate/weather hazards and urban crises, and only those telling real-world crisis events occurring in the study period were retained. Note that even though we searched the news articles with this pre-determined list of hazardous events, the resulting news articles may reveal other types of hazardous events beyond the list. Also, the same article may be identified with different search words. As a result, we have collected a total of 1,941 unique news articles for the 13 types of hazardous events that may correlate with each other (Table 1). For each news article, we collected data including the title, date, full contents, as well as the link to the original webpage.

4. Method

We used a stepped approach to extract the KG, including entities and relationships, from each news article leveraging LLM and RAG techniques. Fig. 1 shows the pipeline of the proposed method. The following subsections introduce each step in Fig. 1 in detail.

4.1. Constructing a knowledge base storing generic triplets

First, we tasked the LLM to generate potential triplets representing the relations between *modulators*, *drivers*, *hazards*, and *impacts*, referring to the definitions of compound events provided by Zscheischler et al.[4]. However, we extend the definition to also include crisis events occurring in urban settings such as urban fires, power outages, and traffic accidents. We prompted to let LLM brainstorm and generate as many triplets as possible, each taking the format of *"[type] source event → relation → [type] target event"*. Then the research team reviews the generated triplets to ensure the syntax is proper and diverse. This step yields 455 triplets depicting the causal or influential relations among different crisis events and their impacts. Note that the triplet generated in the first step only includes very *general* information. For example, it includes economic damages as the impact rather than the specific types of economic damages (e.g., flooded vehicles or reduced crawfish production). Examples of the generated generic triplets include.

Table 1
The pre-determined list of hazardous events for querying news articles.

Event type	Search word (in Chinese)	Number of news articles
High temperature	"高温"	162
Food security	"食物中毒"	26
Security accidents	"事故"	151
Power outage	"停电"	21
Flooding	"内涝"	223
Typhoon	"台风"	430
Rainstorm	"暴雨"	358
Pollution	"污染"	63
Landslide	"滑坡"	34
Urban fire	"火灾"	58
Environmental/facility destruction	"破坏"	28
Strong convection	"强对流"	367
Noise	"噪音"	336

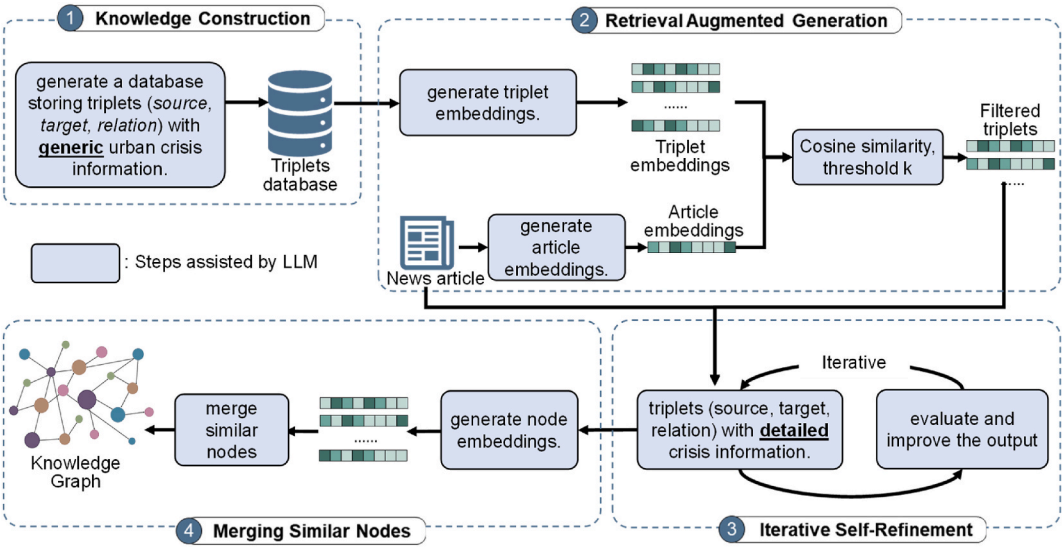


Fig. 1. The pipeline for extracting structured triplets from news articles.

- [hazard] extreme event → result in → [impact] economic damage
- [modulator] climate change → contributes to → [hazard] droughts
- [driver] blizzard → contributes to → [hazard] avalanche

4.2. *Filtering generic triplets based on embedding similarities*

In the second step, we used the embedding function of the LLM to generate embeddings for each triplet. These embeddings are numeric vectors that capture the semantic content of the triplets. Similarly, for each news article, a corresponding embedding was generated with the same function. In the original paper for RAG [37], a similarity measure, i.e., maximum inner product search (MIPS), was used to retrieve top-K relevant documents for further query. In our study, we made slight modifications by adopting the cosine similarity metric and used a threshold, i.e., *k*, to filter *generic* triplets referring to their respective similarity scores. This adjustment considers that some news articles may include more crisis information than others and, hence, should yield more triplets for further queries. The effectiveness of using cosine similarity in this context is supported by findings in recent work such as Lakatos et al. [38] and Juvekar & Purwar [39]. The result of this step is a set of generic triplet embeddings that will be used for subsequent query processing.

The embeddings of the filtered generic triplets were used to retrieve relevant information from the news articles, which yielded triplets with more detailed information. For example, a query with *the generic* triplet *"[hazard] extreme event → result in → [impact] economic damage"* can yield "*[hazard] rainstorm → result in → [impact] inundated vehicles*" with detailed hazard and impact information retrieved from the news article. To improve the quality of the query results and facilitate the following processing, we included the following techniques in the prompt engineering.

1) We provided the definition for the four types of entities referring to Zscheischler et al.[4] and specified the relations for searching, including: a) *exacerbates*: indicates that one entity exacerbates the impact of another; b). *triggers*: indicate that one entity triggers the occurrence of another; c). *weakens*: indicates that one entity weakens the occurrence of another; and d) *enhances*: indicates that one entity enhances the occurrence probabilities of another. LLMs will examine the relationships among entities based on the provided articles with their deductive capabilities.
2) We provided detailed instructions in the prompt to inspire the model to think in a step-by-step manner, following the Chain of Thought (CoT) approach.
3) We requested the output format to be in JSON, with the response including key-value pairs for nodes and edges. These directed edges are used to express the "causal" and "influential" relationships among entities.
4) We asked the model to provide a citation from the supplied news article that substantiates the identified triplet. This can effectively mitigate the hallucination issue present in the LLM.

Readers can check Fig. SI–1 in the supplementary information for the detailed prompt.

4.3. *Refining retrieval results with iterative prompting*

The steps outlined above can effectively identify triplets from news articles, however, they may not always yield complete and

accurate results. To address this limitation, we adapted the "self-refinement" technique described in Madaan et al. [40]. The core idea of their approach is to use LLM as both a "generator" for generating responses and an "evaluator" for providing feedback. Then the LLM uses this feedback to iteratively revise and improve its output. In our method, we implemented this self-refinement process by prompting the LLM to re-read the news article, review the initially extracted triplets, and refine the outputs. Specifically, we instructed the "evaluator" to identify missing entities, remove entities that were not supported by the article, and verify the relationships connecting the different entities. This additional step significantly reduces hallucinations. The refinement process was performed iteratively until one of two stopping conditions was met: 1) the identified entities and relationships remained unchanged between iterations, or 2) the pre-set maximum number of iterations was reached.

Below, we present the pseudo-code for this iterative self-refinement mechanism. Fig. SI–2 in the supplementary information shows the detailed prompt used for self-refinement. Additionally, readers can access the GitHub repository (https://github.com/06chen-xr/kg-llm) for the full list of prompts and the *generic* triplets generated in the first step.

Algorithm 1 Iterative Extraction

Input: news_article (Text), retrieved_knowledge (Text), max_iterations (Integer)

Output: FinalEntityAndRelationList (JSON)

```
function IterativeExtraction(news_article, retrieved_knowledge, max_iterations)
    InitialPrompt ← GenerateInitialPrompt(news_article, retrieved_knowledge)
    InitialResult ← QueryModel(InitialPrompt)

    PrevTotal ← 0
    Total ← 0
    LoopCounter ← 1

while LoopCounter ≤ max_iterations do
        ParsedData ← ParseEntitiesAndRelationships(InitialResult)
        NumEntities ← CountEntities(ParsedData)
        NumRelationships ← CountRelationships(ParsedData)
        Total ← NumEntities + NumRelationships

    if Total ≤ PrevTotal then
            Print("No further improvement, stopping iteration.")
            break

        PrevTotal ← Total
        FollowUpPrompt ← GenerateFollowUpPrompt(ParsedData, news_article)
        FollowUpResult ← QueryModel(FollowUpPrompt)
        InitialResult ← FollowUpResult

        Print("Iteration:", LoopCounter, "Entities:", NumEntities, "Relationships:",
    NumRelationships, "Total:", Total)

        LoopCounter ← LoopCounter + 1

    FinalEntityAndRelationList ← ParseEntitiesAndRelationships(InitialResult)
    return FinalEntityAndRelationList
end function
```

4.4. Merging nodes with similar semantic meanings

With the previous steps, each news article is converted into multiple structured triplets containing *detailed* crisis information, which can be used to construct a KG. However, a challenge here is that different articles may use varying terms to describe the same or similar events, such as "rain storm" and "heavy rain". This lexical diversity results in a bulky and fragmented KG with redundant nodes,

thereby limiting its utility for downstream analysis. To mitigate this issue, we merged entities with similar semantic meanings.

Specifically, we first generated distributed representations for each entity using OpenAI's "*text-embedding-3-small*" model [41], which produces 1536-dimensional embedding vectors that preserve semantic relationships in vector space. We then computed pairwise cosine similarities between all entity embeddings to quantify their semantic similarity. Fig. 2 visualizes a representative subset of entities. Next, we used a graph-based clustering approach to group semantically similar entities into the same group by treating each entity as a node and connecting nodes with edges when their cosine similarity exceeded a pre-defined threshold. This threshold was empirically determined through manual evaluation of numerous term pairs to strike a balance between reducing redundancy and preserving semantic distinctions. The connected components of the resulting graph, representing groups of directly or transitively linked nodes, naturally emerged as semantic clusters. This graph-based clustering approach offers two main advantages: 1) it does not require pre-defined cluster numbers, while 2) it strictly enforces semantic similarity constraints specified by the pre-defined threshold.

Although this automated process effectively consolidated synonymous entities, we recognize that fixed thresholding can occasionally produce erroneous groupings. To minimize such errors, we adopted a relatively high similarity threshold of 0.9 and subsequently conducted manual verification to identify and correct misgroupings. Finally, within each cluster, we selected the most frequently occurring term as the canonical label to ensure consistency across the KG. This step significantly reduced the number of entities from 4,666 to 1,830.

4.5. Method evaluation

4.5.1. Comparison with different prompting techniques

We evaluated our proposed method against three well-established prompting techniques using 50 news articles, including.

- *Few-shot prompt*, which includes a few examples in the prompt to help the LLM understand the task.
- *CoT-prompt* [42], which uses a step-by-step reasoning chain to facilitate progressive thinking in the LLM.
- *ReAct prompt* [43], which applies a multi-step reasoning process, organized in structured stages (e.g., understanding the problem, reasoning, executing tasks, verifying results, and iterating) to improve the output quality.

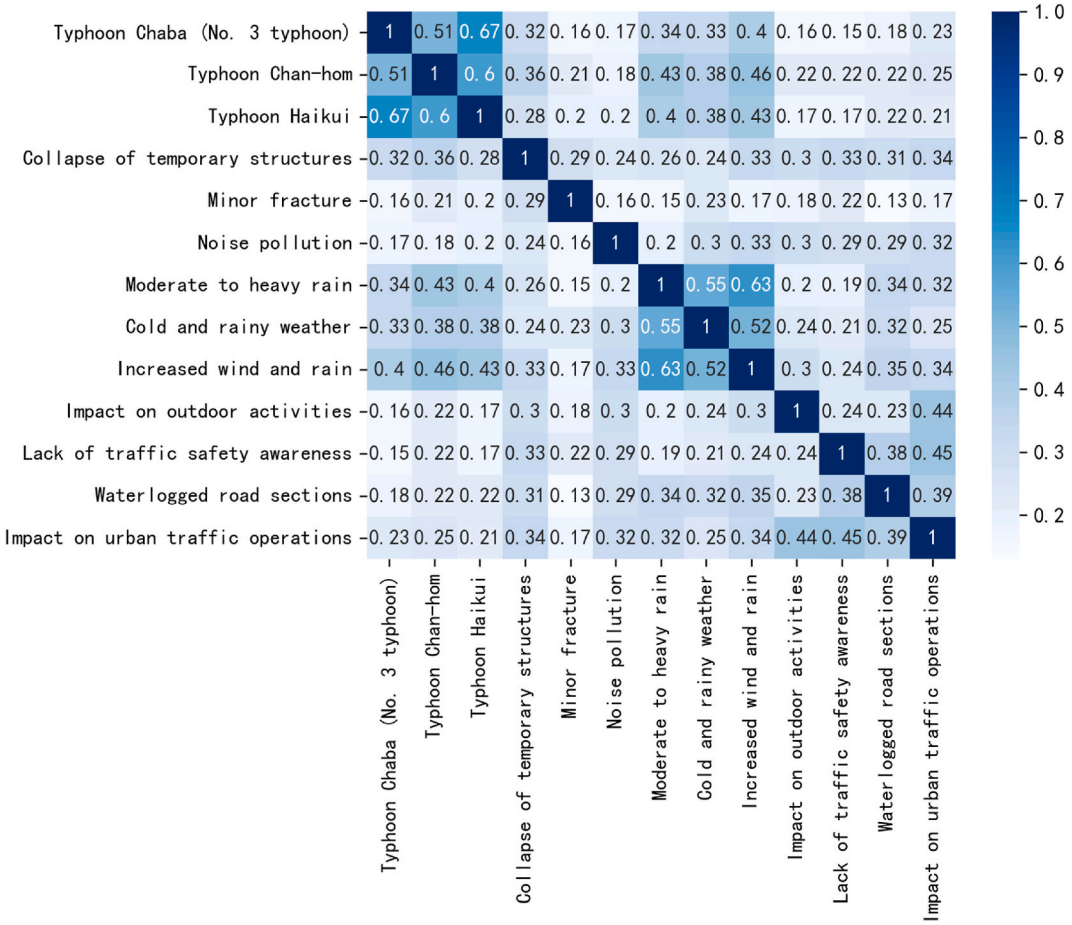


Fig. 2. The semantic similarity matrix of a subset of entities.

To ensure methodological consistency, we maintained the same definitions of entities and relationships across all prompt types and used the latest version of the ChatGPT model, i.e., GPT-4o [44], as our base LLM. Two trained research assistants annotated all articles following a rigorous protocol.

We adopted standard information extraction metrics based on triplet-level matching. A true positive (TP) denotes a triplet identified by both the annotators and the automated method. A false positive (FP) refers to a triplet identified only by the automated method, while a false negative (FN) is identified only by the annotators. For this experiment setting, it is not feasible to determine true negatives (TN), i.e., triplets that are neither identified by annotators nor predicted by the model. Based on the three metrics, we calculated the precision, recall, and F1-score using Equations (1)–(3):

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$F1 - score = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)} \tag{3}$$

In this study, high precision suggests that most extracted triplets are valid and suitable for downstream KG construction, whereas high recall suggests that the method captures most relevant knowledge with minimal omission. The F1-score provides the harmonic mean of precision and recall.

Acknowledging that human annotators may not always outperform LLMs, we adopted a step-wise annotation procedure. For each news article, annotators first read the original text to identify relevant entities and relations based on their own understanding, creating the initial annotation. Second, annotators were presented with the four distinct sets of automatically extracted KGs without information about the underlying models. These outputs served as references for annotators to adjust or enrich the initial annotation, allowing for correction of omissions or ambiguities. Third, for each model-generated output, annotators recorded the number of entities and relations detected, and calculated TP, FP, and FN by comparing the model-generated results and annotation results. A one-hour training session was organized to ensure annotators understood the research context and fundamental concepts (e.g., compound urban crises and knowledge graph). A few standard annotation examples were also presented. Annotators were required to pass a qualification test by annotating two sample articles before beginning the formal task. During the annotation phase, we routinely checked subsets of annotations and provided timely feedback to correct errors and reinforce standards. After finalizing the annotations, we calculated precision, recall, and F1-score for each method by comparing its outputs to the annotations (Table 2).

As shown in Table 2, our method outperforms all baseline prompting strategies (i.e., Few-shot, CoT, and ReAct) in entity recognition by achieving a precision of 0.7515, a recall of 0.6987, and an F1-score of 0.7241. Once entities are correctly identified, it is not difficult for the LLM to infer directional relations between entities using the supporting sentences. While there are no substantial differences in recall among the four methods, indicating that all methods are generally effective at identifying phrases representing key elements of crisis events, our method's significantly higher precision score (i.e., 0.7515) shows that our approach is more effective in filtering out *hallucinated* entities. In our application scenarios, the impact of lower recall can be mitigated by incorporating more data (i.e., news articles) into the KG construction process. Therefore, precision is a more critical metric for model comparison here.

4.5.2. Comparison with conventional naturalistic language processing (NLP) model

Beyond prompting methods, we also evaluated a conventional Natural Language Processing (NLP) model by fine-tuning a pre-trained BERT model, i.e., "*bert-base-chinese*" [45], downloaded from the HuggingFace platform. The model was fine-tuned using our annotated dataset and evaluated on new articles for entity extraction and relation classification. BERT performs token-level classification, for which we adopted the standard BIO tagging scheme—marking the beginning (B), inside (I), and outside (O) of entity spans. However, performance was found limited: the fine-tuned model failed to reliably identify hazard names and frequently mislabeled unrelated words as entity components. The labeling process and an example output from the fine-tuned BERT model are shown in Fig. 3.

4.5.3. Model-agnostic performance evaluation

To assess the generalizability of our method, we compared two LLMs: GPT-4o [44] and DeepSeek. All other settings were kept constant. Results across several test samples showed a high degree of consistency between the two models. Comparatively, DeepSeek demonstrated strong comprehension in Chinese texts. These findings indicate that our framework does not rely on a single proprietary

Table 2
The performance of different prompting techniques for entity recognition.

Method	Precision	Recall	F1-score
Fewshot	0.6676	0.6944	0.6807
CoT	0.6724	0.6714	0.6719
ReAct	0.5660	0.6944	0.6237
Ours	**0.7515**	**0.6987**	**0.7241**

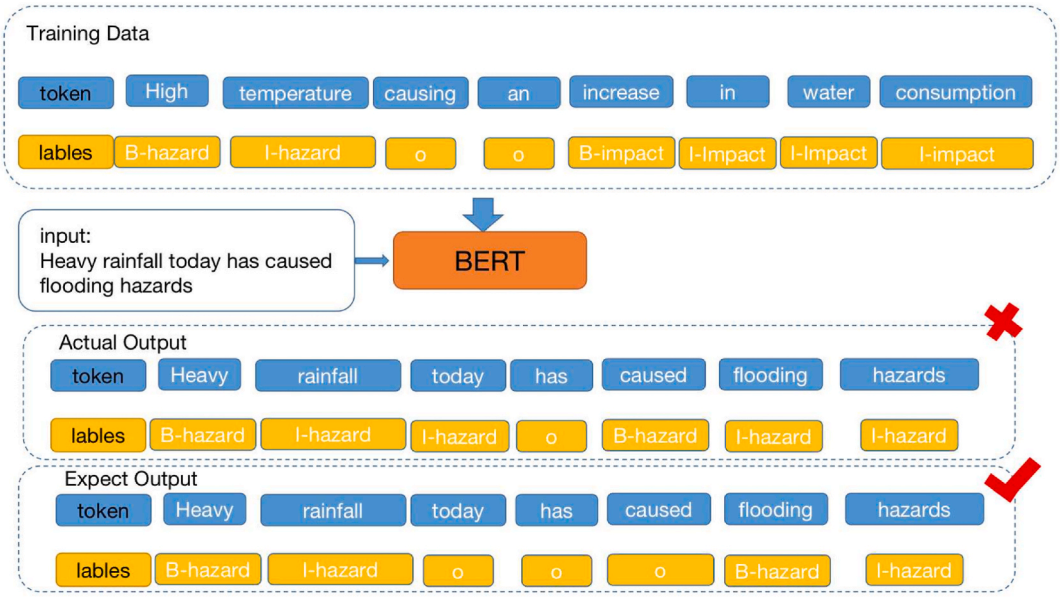


Fig. 3. The Begin-Inside-Output annotation of the BERT model and example output.

model. Its consistent performance can be attributed to the strong instruction-following capabilities of both LLMs, which enable effective execution of our prompt-based and iterative reasoning design. Readers are encouraged to implement the framework using other capable LLMs.

5. Results for complex urban crisis patterns

5.1. *Knowledge graphs for different types of urban crises*

With the developed algorithm, each news article can be transformed into a knowledge graph. Nodes in these graphs represent the four types of entities: *modulator*, *impact*, *driver*, and *hazard*, while edges depict the relationships among them. Fig. 4 illustrates an example of a knowledge graph extracted from a news article. We used different colors to distinguish the various entity types and highlighted the texts where the model finds evidence for the identified entities and relationships (Fig. 4). A single news article can include multiple entities and relationships, forming a small knowledge graph.

With multiple news articles describing the same type of event, such as heatwaves occurring at different times, we can merge their associated knowledge graphs into an integrated graph. This provides a more comprehensive understanding of the event type. Fig. 5 shows an example of an integrated knowledge graph, which is based on news articles retrieved using the keyword “fire [火灾]”. The integrated knowledge graph is visualized with Gephi, a professional graph analysis and visualization software. To simplify the visualization, we do not distinguish between different entity and relationship types. In Fig. 5, the node size is proportional to the node degree, i.e., the number of connections to other entities, while the edge width corresponds to the edge weight (i.e., the frequency of mentions for this relationship). The directional edges indicate the directional relationships, such as exacerbating, enhancing, triggering, and weakening, between the source and target nodes.

Fig. 5 illustrates the complex interplay of factors influencing fires in Shenzhen. Based on Fig. 5, the predominant causes of fires are linked to electric vehicle (EV) batteries, particularly overcharging, spontaneous combustion, and collision, which collectively account for a significant portion of fire incidents. These events often lead to severe consequences, including injuries, fatalities, and property damage. The spontaneous combustion of EVs on roads can cause traffic congestion, further complicating emergency response efforts. Another key finding is the role of human factors in fire accidents. Humans’ social and cultural activities, such as sacrificial rituals and festivals involving burning paper offerings and fireworks, are major contributors to forest fires. These fires, in turn, lead to air pollution and significant environmental damage. Environmental conditions, such as dry weather and strong winds, act as modulators, increasing the likelihood and severity of forest fires. Additionally, unregulated human behaviors such as carelessly discarded cigarette butts, playing with fire, and violations of safety regulations are significant contributors to fire risks. Furthermore, the data reveal that the accumulation of combustible and flammable materials and aging electrical systems in residential buildings are critical risk factors.

The consequences of fires extend beyond immediate damage, with cascading effects such as economic losses, facility closures, and administrative detentions (Fig. 5). Poor management practices, including the lack of firefighting water sources and delayed fire rescue responses, further exacerbate the spread and impact of fires. Systemic issues, such as the perfunctory nature of emergency drills and inadequate safety supervision, were identified as significant barriers to effective fire prevention and response.

Knowledge graphs created for other event types, along with the interpretations, are included in the Supplementary Information.

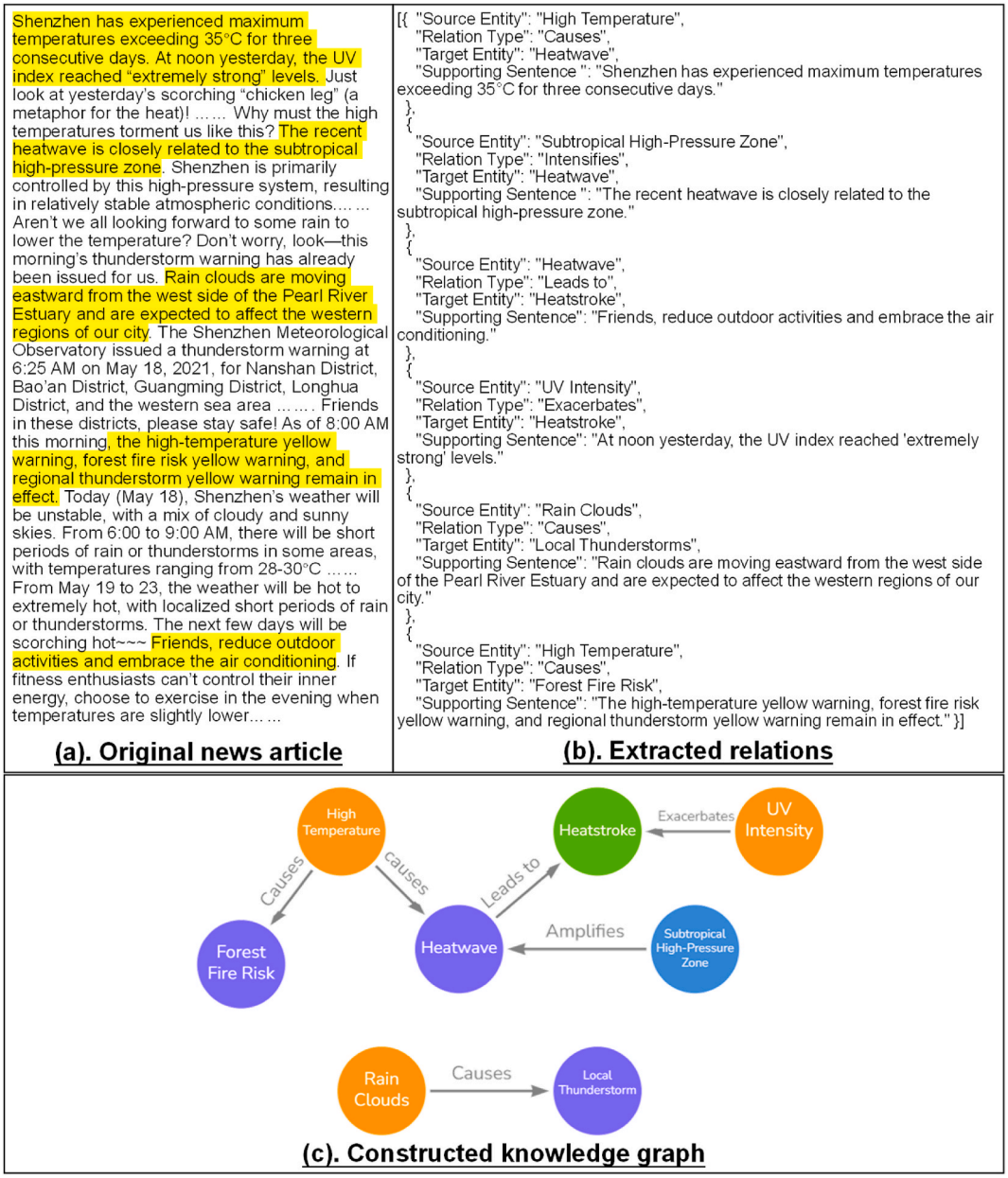


Fig. 4. An example knowledge graph extracted from a news article: a) original news article; b) extracted relations; c) constructed knowledge graph (translated in English).

Table 3 shows the metrics of graphs generated from news articles retrieved using different search words. Events with more mentions in the dataset are associated with a higher number of weighted degrees and average path lengths (Table 3).

5.2. Interpretation of compound urban crises with community detection results

We also created a knowledge graph using all news articles collected for this study, which includes 1 nodes, 3517 unique edges, and with an average weighted degree of 4.90. For clarity, Fig. 6 only shows a subset of the graph with nodes that have a weighted degree of at least three, thus excluding nodes that are only mentioned once or twice in the dataset. Fig. 6 shows that many events are highly interconnected, indicating patterns of compound urban crises. We applied community detection algorithms to the graph. Community detection is a type of algorithm applied to networked data that divides nodes into different communities. Nodes in the same communities are more closely connected to each other and relatively less connected to nodes in other communities. In this way, we can identify events that are closely related, i.e., those mentioned in the same news articles and with causal or influential relationships

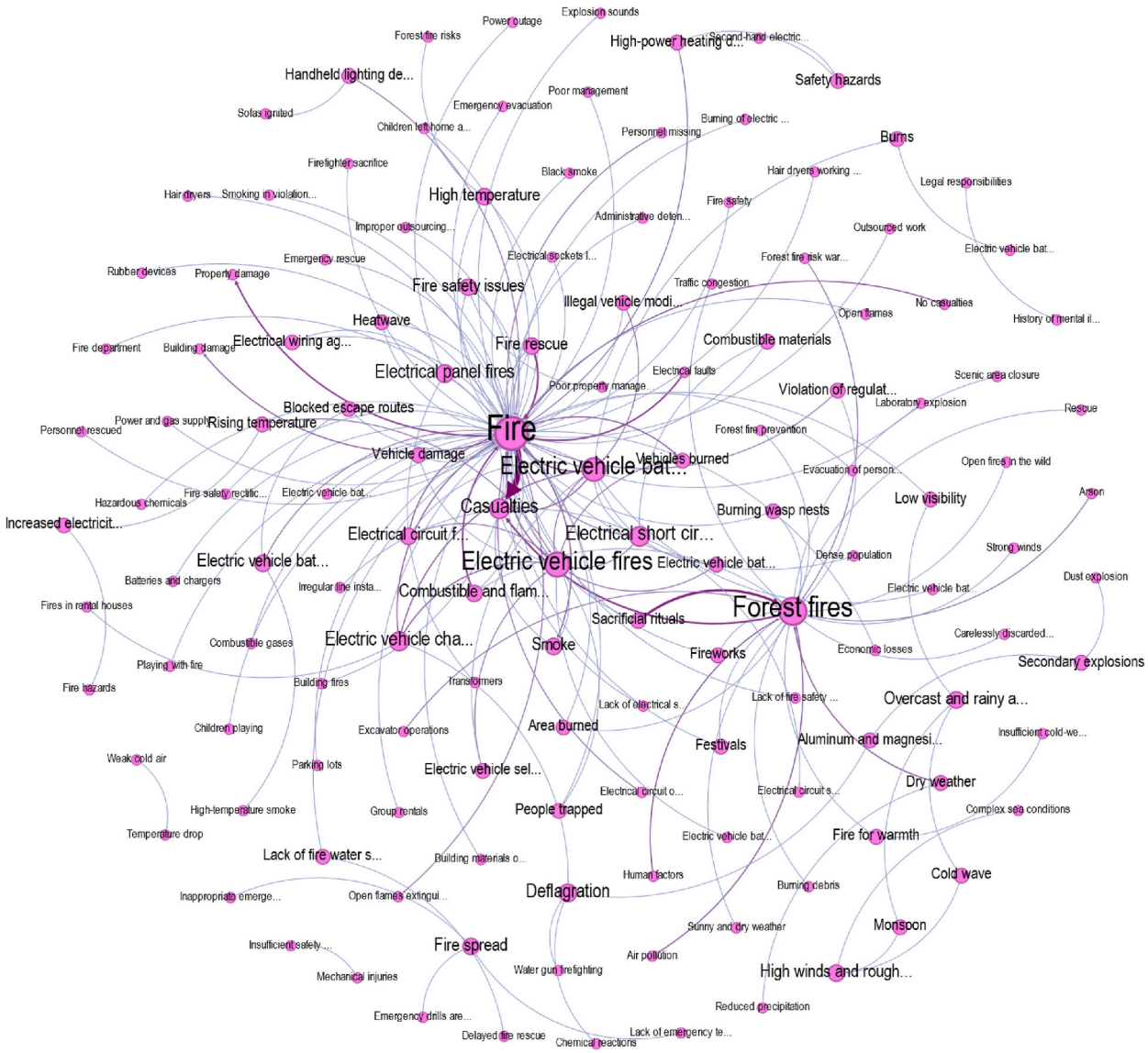
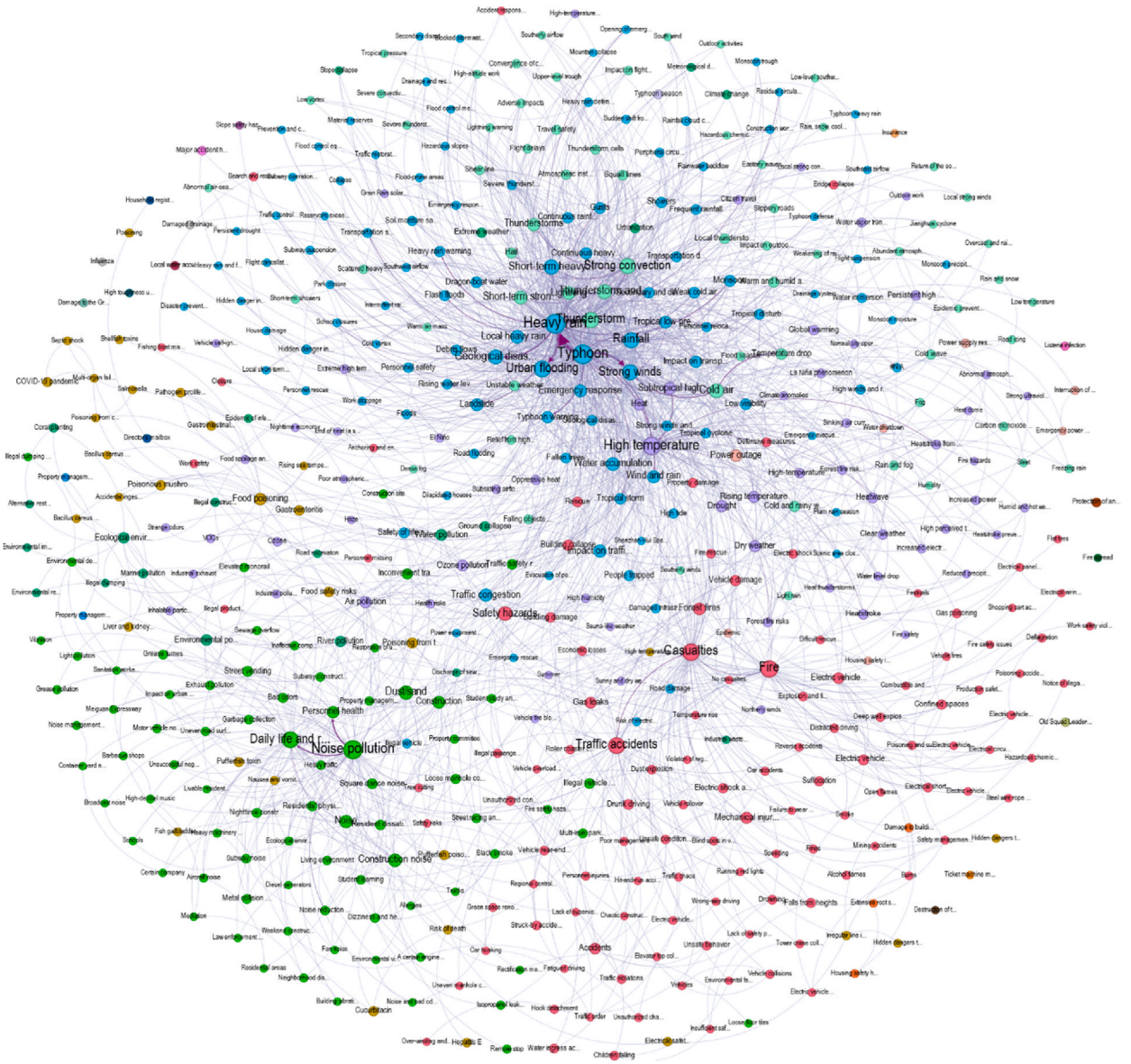


Fig. 5. The integrated knowledge graph extracted with news articles retrieved using the keyword "fire":(translated in English).

Table 3
The metrics of knowledge graphs constructed for different types of events.

Search word	# of nodes	# of unique edges	Average weighted degree	Network diameter	Graph density	Average path length
High temperature	250	401	3.096	7	0.006	2.851
Pollution	144	183	1.715	5	0.009	2.236
Fire	138	167	1.21	6	0.009	2.55
Food security	102	120	1.265	4	0.012	1.839
Safety accidents	414	504	1.754	6	0.003	2.471
Power outage	83	99	1.53	5	0.015	2.201
Flooding	308	567	4.477	8	0.006	2.874
Typhoon	384	727	6.195	7	0.005	3.068
Rainstorm	361	759	5.615	7	0.006	2.708
Landslide	115	141	1.852	4	0.011	2.134
Destruction	122	116	0.975	5	0.008	1.608
Strong convection	341	747	6.774	8	0.006	3.075
Noise	467	635	2.381	8	0.003	2.873


Heavy rain
Thunderstorm
Rainfall
Typhoon
Strong winds
Urban flooding
Geological disas.
Short-term heavy
Strong convection
Thunderstorms
Local heavy rain
Emergency response
Subtropical high
Cold air
Landslide
High temperature
Power outage
Wind and rain
Water accumulation
Heatwave
Drought
Rising temperature
Food poisoning
Water pollution
Ozone pollution
Traffic congestion
Air pollution
Safety hazards
Casualties
Fire
Electric vehicle
Traffic accidents
Noise pollution
Daily life and t...
Dust and
Construction
Gas leaks
Personal health
Construction noise
Accidents
Mechanical injur...
Suffocation
Electric shock a...
Drunk driving
Vehicle rollover
Illegal vehicle
Dust exposition
Dust storms
Frequent rainfall
Typhoon season
Heavy rain warning
Extreme weather
Transportation s...
Transportation d...
Impact on transp...
Temperature drop
Cold wave
Low visibility
Tropical cyclone
Tropical low
Typhoon Wutip
Typhoon Yagi
Rising water lev...
Unstable weather
Flash floods
Dragon boat water
Short-term stron...
Thunderstorm and...
Warm and humid a...
Persistent high ...
Global warming
Monsoon trough
Monsoon precipit...
Rain and snow
Rain and fog
Fog
Haze
Dry weather
Forest fires
Forest fire risk
Fire hazards
Heatstroke
Gas poisoning
Fire safety
Fire safety issues
Vehicle fires
Electric vehicle...
Electric short...
Electric panel...
Electric wirin...
Fireproof
Fire hydrant
Fire brigade
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identified by LLM, which may suggest the common patterns of compound events in the study city.

Specifically, we used the Louvain algorithm for community detection due to its high efficiency [46]. The algorithm identified 65 communities, many of which are small and only include a few nodes. We focused on the eight communities with more than 6 nodes, of which the eight communities jointly include 1659 (or 90.66 %) nodes and 3377 (or 96.02 %) unique edges. Nodes in different communities are represented by different colors in Fig. 6. The following subsections describe the compound urban crises identified based on the eight communities.

5.2.1. Community 1: noise and air pollution near construction sites

Community 1 focuses on *noise pollution* and its multifaceted impacts on urban life (Fig. 7). A compound pattern identified is that *noise pollution* significantly disrupts human activities such as *student learning* (weight = 5) and *residents' daily life and rest* (weight = 98), with severe consequences for *personnel health* (weight = 105) and *residents' physical and mental well-being* (weight = 16). Primary sources of noise include *construction activities* (weight = 10) (e.g., *heavy machinery*, weight = 2; *nighttime construction*, weight = 5), *roadside businesses* (weight = 3), *garbage collection* (weight = 5) in the early morning, *square dancing* (weight = 4) in the evening, among other types of human activities. Built-environment factors such as *loose manhole covers* (weight = 5) and *uneven road surfaces* (weight = 2) are reported to exacerbate noise generated from *heavy traffic* (weight = 2), while *subway lines* and *illegal vehicle modifications* (weight = 3) further compound the issue.

A critical compound event revealed in this community is the dual impact of *construction sites*, which generate not only *noise pollution* (weight = 4) but also *dust/sand pollution* (weight = 5). This combination leads to *air pollution* and *respiratory health risks* (weight = 6), while persistent noise causes *mental stress* (weight = 2) and sleep deprivation. The cascading effects extend to urban governance challenges such as *residents' dissatisfaction* (weight = 4) and *ineffective complaint feedback* (weight = 4), highlighting the need for systemic mitigation measures to address noise pollution challenges.

5.2.2. Community 2: strong winds, heavy rainfall, and their cascading impacts during typhoon season

Community 2 highlights a cluster of events related to *heavy rainfall* and *typhoons* (Fig. 8). A key compound pattern identified in this community is that *heavy rainfall* (weight = 28), often intensified by *monsoons* (weight = 10) and *typhoons* (weight = 413), leads to *urban flooding* (weight = 217), which further *disrupts transportation systems* (weight = 32) and *damaged infrastructure* (weight = 3). Specifically, *road flooding* (weight = 24) and *fallen trees* (weight = 13) exacerbate transportation challenges, while *low visibility* (weight = 6) during adverse weather conditions increases the risk of safety accidents.

The community also reveals the role of *typhoons* in amplifying the impacts of *heavy rainfall*. *Typhoons* bring *continuous rainfall* (weight = 6), which *saturates the soil* (weight = 8) and causes *landslides* (weight = 58), *debris flows* (weight = 31), *flash floods* (weight = 45), and other *geological disasters* (weight = 131). These events pose significant risks to *personnel safety* (weight = 3) and often necessitate *emergency evacuations* (weight = 4) and the opening of *emergency shelters* (weight = 3). Additionally, *typhoons* can lead to *school closures* (weight = 12), *flight cancellations* (weight = 12), and *transportation suspensions* (weight = 28), disrupting city residents' daily lives and economic activities.

5.2.3. Community 3: urban accidents and their cascading impacts

Community 3 focuses on events associated with injuries and fatalities, including *urban fires*, *transportation accidents*, and *industrial safety incidents* (Fig. 9). A compound crisis pattern identified is that electric vehicles (EVs) are a major source of urban fires, driven by *overcharging* (weight = 6), *battery explosions* (weight = 3), and *self-ignition* (weight = 3). These incidents often escalate into fires that result in *casualties* (weight = 32) and *property damage* (weight = 5). *Transportation accidents* are another critical focus in this community, with *traffic accidents* (weight = 43) linked to high casualty rates. Unregulated human behaviors, including *drunk driving* (weight = 7), *distracted driving* (weight = 5), *vehicle overloading* (weight = 2), and *running red lights* (weight = 10), are the major contributors. Human errors, such as *blind spots in vision* (weight = 10) and *weak traffic safety awareness,* further amplify the risks of fatal traffic accidents.

Industrial and construction accidents compound these risks. Poor management practices, such as *violation of regulations in operations* (weight = 3), *poor supervision* (weight = 2), and *safety management deficiencies* (weight = 2), lead to incidents like *mechanical injuries* (weight = 7) and *falls from height* (weight = 6). A critical compound event in industrial factories involves *combustible materials* (weight = 5) igniting fires, which can trigger *explosions* (weight = 4) and secondary hazards like *toxic smoke* (weight = 3), resulting in multifaceted crises.

5.2.4. Community 4: high temperatures and their cascading impacts

Community 4 highlights several compound urban crises related to *high temperatures*, driven primarily by the *subtropical high-pressure*

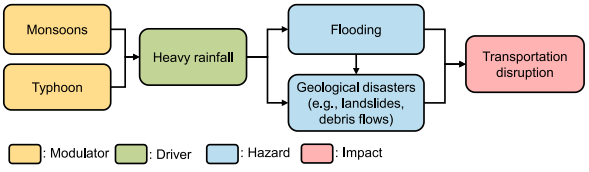


Fig. 8. A major compound urban crisis identified in Community 2.

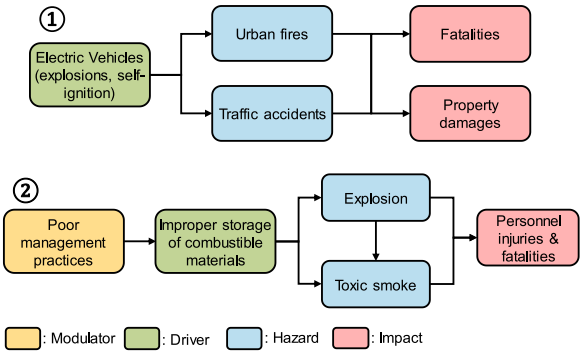


Fig. 9. Major compound urban crises identified in Community 3.

system (weight = 69) (Fig. 10). A key compound pattern identified in this community is that *high temperatures*, particularly during dry seasons like autumn, heighten the risk of *forest fires* (weight = 7). Dry branches and leaves are more likely to catch fire, especially in Shenzhen, a city known for its high vegetation rates, leading to increased *forest fire risks* (weight = 18). These fires not only damage ecosystems but also contribute to *air pollution* (weight = 3).

High temperatures are also closely associated with heat-related diseases, such as *heatstroke* (weight = 17), particularly during periods of *oppressive heat* (weight = 2) and *heatwaves* (weight = 16). In spring and early summer, *high temperatures* combined with *high humidity* accelerate food spoilage (weight = 2) and increase the risk of food safety accidents such as *food poisoning* (weight = 4). These conditions also create favorable environments for the growth of *poisonous mushrooms* (weight = 3), which can lead to severe health issues, including *gastroenteritis* (weight = 2) and *multi-organ failure* (weight = 2).

Additionally, *high temperatures* exacerbate *ozone pollution* (weight = 9) by increasing the levels of *volatile organic compounds (VOCs)* (weight = 3) and *nitrogen oxides* (weight = 2). This is particularly significant in warm cities like Shenzhen. The resulting *ozone pollution* poses serious *health risks* (weight = 7), including *respiratory diseases* (weight = 3). Environmental factors, such as *dry weather* (weight = 7) and *reduced precipitation* (weight = 3), can amplify the impacts of *high temperatures* on *forest fires* (weight = 7) and *air pollution* (weight = 3).

For urban operations, *high temperatures* increase the *power grid load* (weight = 15), posing challenges to *energy infrastructure* (weight = 2). They also increase the likelihood of *vehicle tire blowouts* (weight = 3) and *vehicle self-ignition* (weight = 4), which can disrupt urban transportation systems.

5.2.5. *Community 5: strong convection and severe weather events*

Community 5 focuses on events related to *strong convection* and its cascading impacts on urban systems (Fig. 11). A compound crisis pattern identified in this community is that *cold air* leads to *temperature drops* (weight = 67), while the *convergence of cold and warm air currents* (weight = 3) generates *strong convection* (weight = 16). This *strong convection*, in turn, triggers severe weather phenomena such as *thunderstorms* (weight = 45), *lightning* (weight = 18), *short-term strong winds* (weight = 30), and *hail* (weight = 38). These events can cause drastic weather changes, such as sudden transitions from drought to flood, imposing significant stress on cities' emergency management efforts.

Environmental factors, such as the *rainy season*, further amplify the impacts of *strong convection*. For instance, *cold air* (weight = 2) and *warm, humid air currents* (weight = 6) interact to intensify atmospheric instability, leading to more frequent and severe weather events. The consequences of these events range from *property damage* and *economic losses* to *travel disruptions*. For example, *short-term*

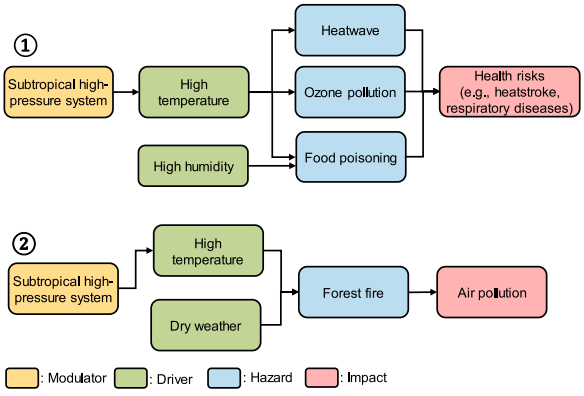


Fig. 10. Major compound urban crises identified in Community 4.

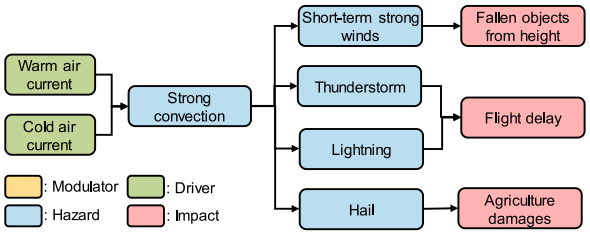


Fig. 11. Major compound urban crises identified in Community 5.

strong winds and *thunderstorms* can cause *falling objects from heights* (weight = 4), posing risks to public safety, while *lightning* and *thunderstorms* can lead to *flight delays* (weight = 2) and disrupt *travel safety* (weight = 2). More severe events, such as *tornadoes* (weight = 4) and *hailstorms*, can cause significant damage to urban infrastructure and agriculture.

5.2.6. Community 6: safety hazards from improper construction activities

Community 6 focuses on safety hazards arising from improper or illegal construction activities (Fig. 12). A compound pattern identified in this community is that irregular construction practices, such as *unauthorized construction* and *unauthorized connection of natural gas pipelines*, can damage critical infrastructure and cause *gas leaks* (weight = 2) or *water ingress accidents* (weight = 4). These incidents pose significant risks, including *suffocation* (weight = 2) and *personnel fatalities* (weight = 1). Environmental factors, such as the *rainy season*, increase the likelihood of *water ingress accidents* (weight = 3) and destabilize construction sites and surrounding areas. Additionally, *chaotic construction management* (weight = 1) and *poor enterprise management* (weight = 1) contribute to unsafe conditions, such as *deep well explosions* (weight = 2), creating a cascade of safety hazards.

The consequences of these safety events include *fines* (weight = 1), *administrative detention* (weight = 1), and *criminal detention* (weight = 1) for construction companies, underscoring the severe human and economic costs of inadequate safety measures. Efforts to address these risks, such as *hidden danger investigation and rectification* (weight = 1) and establishing an *emergency command center* (weight = 1) are discussed.

5.2.7. Community 7: environmental pollution and its cascading impacts

Community 7 focuses on environmental pollution, particularly water pollution caused by illegal wastewater discharge and ecological damage (Fig. 13). A key compound pattern identified in this community is the link between *water pollution* (weight = 3) and the *spread of infectious diseases* (weight = 1), highlighting how environmental degradation can exacerbate public health risks. For instance, *illegal dumping* (weight = 2) and *industrial wastewater discharge* (weight = 4) are major contributors to *river pollution* (weight = 2) and *marine pollution* (weight = 1), which contaminate water sources and create breeding grounds for pathogens.

The community also reveals the role of urbanization in amplifying environmental challenges. Urban projects, such as *road excavation* (weight = 1) and *sewage discharge* (weight = 1), contribute to *muddy water flows* (weight = 1) that pollute rivers and strain sewage systems. Additionally, *dense population areas* (weight = 1) are more vulnerable to the impacts of water pollution, as limited resources and poor waste management systems strain urban ecosystems.

Efforts to address these challenges include environmental improvement initiatives, such as *coral planting* (weight = 1) and *alternative restoration measures* (weight = 1), which aim to mitigate ecological damage. Shenzhen has also implemented policies requiring companies responsible for pollution to replace fines with restoration efforts such as coral planting.

5.2.8. Community 8: power outage and its cascading impacts

Community 8 highlights a cluster of events centered around *power outages* and their cascading impacts on urban systems. *Power outages* in residential areas are often triggered by *housing safety issues*, such as *arbitrary wiring* (weight = 1) and *low-voltage short circuits* (weight = 1), as well as environmental stressors like *substation flooding* (weight = 1) (Fig. 14). A key compound crisis pattern revealed in this community is that *substation flooding* leads to power outages due to *damaged equipment* (weight = 1), further disrupting the operation of other systems, such as *water management systems.* For instance, *water pumps* (weight = 1) require power to function, and their failure during flooding creates a feedback loop of failure and exacerbates urban flooding.

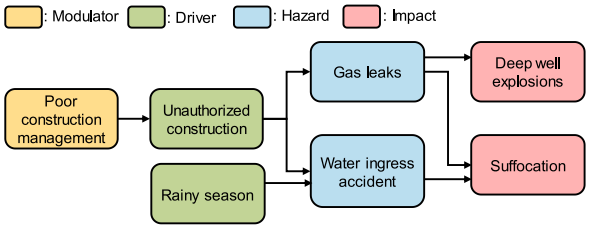


Fig. 12. Major compound urban crises identified in Community 6.

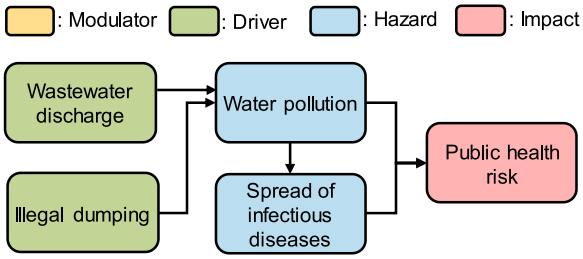


Fig. 13. Major compound urban crises identified in Community 7.

The community also demonstrates how pandemics can strain urban infrastructure, including power supply systems. For example, the increased demand for electricity during a pandemic, coupled with reduced maintenance capacity, exacerbates vulnerabilities, as seen in the link between *epidemics* and *power outages* (weight = 1). Additionally, social factors play a significant role in compounding urban crises. A notable example is *"property bullying"* (weight = 2), where residential managers may intentionally cut off water and power supply due to conflicts or mismanagement.

6. Discussion

In this research, we present a method for understanding compound urban crises by automatically constructing knowledge graphs (KGs) from historical event records. The method's effectiveness was validated using 1941 news articles collected from Shenzhen, China, revealing unique patterns of interconnected urban crises. The key contributions of this study are summarized as follows:

First, while the concept of compound events has been extensively studied in the context of climate and weather (e.g., Ref. [4]), this research expands the framework to urban crises, incorporating a broader range of events such as pollution, security accidents, food poisoning, and power outages. This adaptation is significant because urban crises often arise from the interplay of social, environmental, and built-environment factors, creating complex, cascading impacts that are not adequately captured by traditional models. Crucially, our approach overcomes limitations of purely generative AI models by grounding knowledge in documented local events through RAG, ensuring place-specific understanding of urban contexts that generic LLM outputs cannot replicate. For example, we identified a featured compound pattern in Shenzhen involving high temperatures, ozone pollution, and food poisoning during early summer—a scenario that has not been widely documented in existing literature. Additionally, the uses of graph structure enable linking disparate events across time and sources, particularly valuable for analyzing compound crises that may span months or years. Figs. 7–14 further delineate the varied types of compound urban crises identified with graph analysis. By applying the compound event framework to the urban scale, this study provides a more holistic understanding of how crises interact and amplify one another in cities, offering critical insights for city- and community-level crisis management.

Second, this research advances automated knowledge graph construction by addressing limitations of existing approaches. While recent studies have explored KGs for analyzing complex events and supporting emergency decision-making (e.g., Ref. [1,14,21,24]), many rely on structured datasets or require labor-intensive annotation. Our framework combines the deductive capabilities of LLMs with the structured representation of KGs and hence reduces annotation costs while maintaining accuracy. Key innovations include the use of RAG to minimize LLM hallucinations and an iterative self-refinement mechanism that improves KG quality by prompting the model to verify and revise its outputs. The proposed framework is adaptable to other unstructured data sources, such as government guidelines [21], social media posts (e.g., Ref. [24,30]), and real-time data streams like public service requests, allowing for dynamic knowledge updates during crisis events [34].

Third, the visual and computational properties of KGs, generated by our method, can further support various downstream applications supporting urban crisis management [1,20]. These include fine-tuning domain-specific AI models for improved predictive accuracy, being deployed in a real-time decision-making system to support emergency management activities, or developing question-answering systems to address situational queries from citizens during crises. Furthermore, the generated KGs can help identify complex and challenging compound crisis scenarios for desk simulations, thereby enhancing the preparedness and response capacities of local governments to handle such scenarios [47]. In particular, users can view the obtained KGs as independent and updatable knowledge bases, which can be effectively integrated with different LLMs without repetitive retraining.

Despite these contributions, our research has several limitations that provide opportunities for future work. First, reliance on news articles limits timeliness, as these documents often reflect post-event reporting. Future studies should consider incorporating heterogeneous data sources, such as sensor measurements or citizen-generated reports, to capture real-time dynamics and reduce source-related biases (e.g., Ref. [20]). Particularly, considering that the varied types of urban crises are in fact spatially-referenced, future studies may consider automatic approaches for developing *geographic knowledge graphs* to further uncover the spatio-temporal relations among the events [48]. Second, while we merged nodes using embedding similarity and manual checks, establishing a predefined ontology for urban crises (e.g., Ref. [20]) could improve the integration accuracy. Third, some recent studies have proposed methods for automatically evaluating the outputs of KG and RAG using LLM (e.g., Ref. [49,50]). These methods could be seamlessly integrated into the self-improvement mechanism of our KG extraction process to further improve the quality of knowledge extraction. Fourth, while our method reduces hallucination through RAG and prompting LLMs to cite supporting sentences from original articles, we

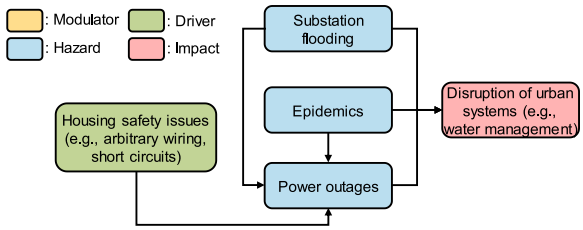


Fig. 14. Major compound urban crises identified in Community 8.

acknowledge that no LLM-based approach can fully eliminate this risk. In crisis-sensitive applications, we recommend combining automated pipelines with expert validation to ensure reliability.

7. Conclusion

Compound urban crises pose significant challenges to cities due to their complex interactions and potential cascading impacts, making them difficult to analyze and manage effectively. This study proposed a method for automatically constructing KGs from historical textual event records to better understand such crises. The method was validated using news articles collected from Shenzhen, China. This study contributes methodologically by introducing a cost-efficient approach that leverages LLM and RAG to construct high-quality KGs without annotation. It also advances urban crisis research by adapting compound event typologies to urban contexts and uncovering location-specific crisis patterns that are critical for city-level emergency management. The resulting KGs offer actionable insights for enhancing emergency decision-making, designing desk simulations, and improving local government preparedness for compound crises. This research highlights the potential of integrating AI-driven approaches like KG into urban management, offering valuable pathways for practitioners and researchers to collaboratively build more resilient and adaptive cities in the face of complex and evolving challenges.

CRediT authorship contribution statement

Haiyan Hao: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xiaorui Chen:** Writing – original draft, Visualization, Validation, Formal analysis. **Yudi Chen:** Writing – review & editing, Project administration, Funding acquisition. **Nan Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (Grant No. 72404236, 72301133) and the University Development Fund (UDF01003238) provided by the Chinese University of Hong Kong, Shenzhen. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Natural Foundation of China and the Chinese University of Hong Kong, Shenzhen.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijdrr.2025.105669.

Data availability

Data will be made available on request.

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